CulinaryState-RAG: A State-Tracking, Process-Aware and Evidence-Abstaining RAG for Context-Aware Cooking, Grounded in a Sparsity-Aware Process-Effect Ledger

Mini-Project Proposal Brief | **Group 17 (MDMA)**: Kush Tokas (2023296), Parv Goyal (2023374) | Computational Gastronomy, Prof. Ganesh Bagler, IIIT Delhi

1. The gap this project closes

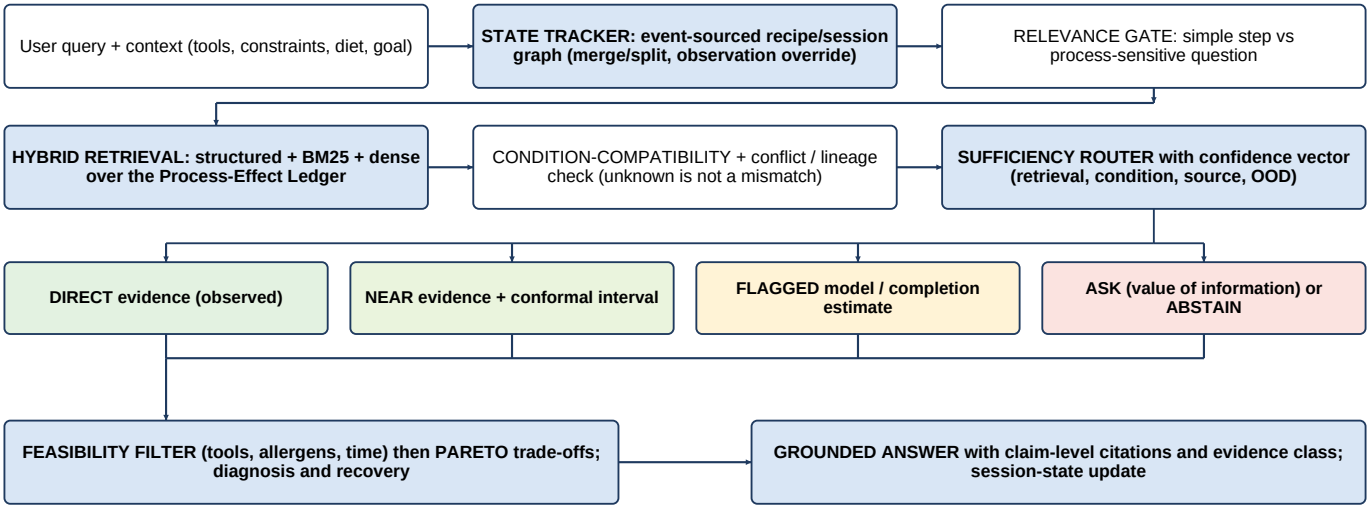
Cooking assistants built on an LLM or plain RAG treat a recipe as text. They do not track what state the food is in right now, do not know what a process does under given conditions (temperature, time, medium), ignore the cook's real constraints (no oven, allergens, time) and answer confidently even when no evidence exists. The scientific evidence that would fix this — how processes change retention, browning and flavor — is **sparse, heterogeneous and confounded**: a systematic analysis of Phenol-Explorer data found the food itself was often a stronger determinant of retention than the process applied. We therefore treat sparsity as a first-class property: we model what has and has not been measured, never invent chemistry, and let the system abstain.

2. Central question and hypotheses

Can explicit food-state tracking plus evidence-bounded process-effect reasoning improve context-aware culinary guidance over text-only and dense RAG — and can the system know when it does not know?

- **H1** State-aware retrieval beats text-only / dense RAG on current-state and next-action questions.
- **H2** Process information beats food-identity-only models on unseen ingredient×process pairs and unseen studies.
- **H3** Evidence-class routing with calibrated abstention lowers the unsupported-claim rate at a small cost in coverage.
- **H4** Feasibility filtering and recovery reasoning respect user constraints and state divergence better than a generic chatbot.

3. System architecture (all decisions by structured modules; the LLM only verbalises and proposes)



4. Deliverables (each unique, non-trivial and non-overlapping with the assignments)

#	Deliverable	Concrete output and success check
D1	**Process-Effect Evidence Ledger + data card**	Versioned table: ≥1,000 rows from public retention databases (Phenol-Explorer, USDA) + ≥40 hand-verified literature rows (raw-vs-cooked, kinetics). Fields: ingredient, process, conditions with status (measured / range / qualitative / unknown), target, value or direction, unit, evidence tier, DOI, censoring flag. Names normalised to RecipeDB/FlavorDB. Check: every row has source + tier; coverage and bias report.
D2	**Sparsity-aware completion and cold-split benchmark**	Quantify density and missing-not-at-random structure; pair-holdout, ingredient-disjoint and study-disjoint splits; baselines (persistence, process-mean, nearest-evidence, LightGBM) vs. biased matrix factorization / factorization machine with side information; propensity-weighted evaluation; split-conformal intervals. Check: answers H2 with study-level bootstrap CIs, negative results reported.
D3	**State-tracking engine (event-sourced recipe/session graph)**	Typed operations (merge, split, reserve, recombine), belief-state fields with provenance, and direct-observation override ('the onions are still pale'). Consumes structured recipe tables / upstream parser output; does not re-implement scraping or NER. Evaluated on 40 gold recipes and 30 scripted sessions: state-field accuracy, replay consistency, merge/split F1.
D4	**Evidence-aware retrieval with calibrated abstention**	Hybrid structured + BM25 + dense retrieval over the ledger; condition-compatibility scoring (unknown is not a mismatch); evidence classes observed / near / interpolated / model-estimated / qualitative / conflict / unsupported; confidence vector; answer-ask-abstain policy. Check: condition-compatible Recall@K, risk–coverage, unsupported-claim rate, ECE.
D5	**End-to-end CulinaryState-RAG + benchmark + demo**	Relevance gate, context-aware feasibility filter, diagnosis and recovery ('accidentally boiled instead of browned') with Pareto trade-offs. Benchmark of ~80 questions over 8 task types, baseline ladder B0 LLM-only, B1 dense RAG, B2 hybrid RAG, B3 structured RAG, B4 state-aware RAG, B5 + evidence routing/abstention; one-at-a-time ablations; oracle-upstream tests; small chef/peer review; 8-scenario Streamlit demo.

5. Resources and API access requested from Prof. Bagler / CoSyLab (IIIT Delhi) and Foodoscope

The project is built on the CoSyLab and Foodoscope data ecosystem. We formally request research access to **all** of the following, at research-level quota, together with whatever further endpoints our API audit shows we need (we will document exactly what we use).

Resource	What we request	Used for
RecipeDB (every current version)	Full API keys plus a bulk snapshot: recipes, ingredients, the process vocabulary, utensils, step-level instructions, ingredient form/state, durations/temperatures where annotated.	State-tracking gold set, process vocabulary, recipe-level context (D3, D5)
FlavorDB / FlavorDB2	Ingredient→molecule mappings, flavor and odor descriptors, SMILES and descriptors, source references.	Evidence linking, FlavorTwin slice (D1, stretch)
Foodoscope API suite	All endpoints: flavor compatibility, nutritional breakdown, ingredient substitution, plus any process, utensil or food-processing endpoints.	Baselines, feasibility checks, RAG tools (D4, D5)
DietRx, SpiceRx, SustainableFoodDB	Read access to health, spice and sustainability associations.	Context and constraint metadata only; no health claims (D5)
IDs and documentation	Cross-resource ID mappings (ingredient, process, molecule), synonym tables, schemas, rate limits, pagination and authentication details.	Entity alignment and API audit (D1, D3)
Unpublished / internal fields	Any process-effect, before/after-state, condition and provenance annotations; existing student work we must not duplicate.	Ledger seeding and non-overlap (D1, D2)
Terms and support	Permission to store derived, non-raw benchmark data; sandbox or staging keys; a named contact in the lab.	Reproducible release of our benchmark

Our commitments: full attribution, strict rate-limit compliance, no redistribution of raw data, and sharing the derived evidence ledger and benchmark back with the lab.

6. How process effects and sparsity become the advantage

- **Process effects:** each answer is conditioned on state, process and conditions, using food-science structure (first-order/Arrhenius exposure features, moisture-gated reactions) as inductive bias rather than a label.
- **Sparsity as signal:** which food×process×target cells were ever studied is informative. We borrow recommender-system tools (low-rank completion, side information for cold start, propensity correction for reporting bias) and keep completed values in a separate, flagged evidence tier that can never enter test or calibration sets.
- **Selectivity:** direct evidence outranks model completion; where neither supports the query the system says so, and asks one high-value question instead of guessing.

7. Domains covered

Domain	What the project does
Food science	Process/condition schema, kinetics-inspired features, retention vs. degradation vs. leaching, honest limits of chemistry claims.
Data engineering	Provenance, evidence tiers, unit/yield handling, censored values (never coded as zero), versioned releases, duplicate-safe splits.
Machine learning	Sparse-data baselines before models, matrix/factor completion, propensity weighting, conformal prediction, selective prediction.
Systems / RAG	State memory, hybrid retrieval, evidence compatibility, structured reasoning packet, claim-level citations.
HCI / user context	Tools, time and dietary constraints as hard filters; one clarifying question only when it changes the decision.
Evaluation	Baseline ladder, ablations, oracle cascades, paraphrase/unit/state/constraint robustness tests, per-frequency error analysis.
Safety and ethics	Allergens and food safety as hard constraints; no medical or health claims; chemical similarity is never presented as sensory equivalence.
Reproducibility	Frozen schemas, fixed seeds, data card, code and demo released with the report.

8. Evaluation, feasibility and risks

- **Headline evaluation:** study-disjoint and pair-holdout splits (random splits only as a sanity check); study-level bootstrap CIs; errors stratified by ingredient frequency.
- **Feasibility:** open data, Python, pandas, scikit-learn, LightGBM, a small retrieval stack and Streamlit; no GPU required; D5 is scoped to ~80 questions so two members can finish it.
- **Risks and mitigations:** sparse or inconsistent units → explicit unit/yield handling and tiers; food identity dominates process → identity-only baseline and pair-holdout expose it; API access delayed → public downloads and web tables as fallback; weak model gains → report the negative result and keep the retrieval-first design.
- **Stretch (if time):** process-conditioned FlavorTwin slice for ≤10 anchor ingredients comparing static vs. cooked flavor-profile neighbours.

Indicative phases (to align with the course calendar): P1 schema + ledger v1 + state-graph spec (D1, D3 start) → P2 splits, baselines, completion (D2) and state tracker (D3) → P3 retrieval + abstention (D4) → P4 integration, benchmark, ablations, demo (D5) → P5 report and code release.

References

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